

Victor Yang

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PERSONAL SUMMARY

I am a 3rd year manufacturing engineering student at UBC and seeking a co-op opportunity. With a proven track record in industry-partnered research and multidisciplinary leadership roles, I've developed strong problem-solving, leadership, and teamwork skills. I'm eager to apply both my technical and collaborative knowledge in a real-world professional engineering environment where I can continue to grow and deepen my understanding about the industry.

TECHNICAL SKILLS

Engineering / Data: Python (NumPy, SciPy, Pandas, Open3D, TensorFlow) | C++ | R | Java | IEC 61131-3
Web Development: HTML | CSS | React | Tailwind | Typescript | JavaScript | Next.js
Software Packages: SolidWorks (CSWA, CSWP) | Siemens NX | Microsoft Office | Gantt Project | Git | Arena Simulation
Manufacturing: Soldering | Hand/Power Tools | Microcontrollers | 3D printing | Metrology | CNC machining | GD&T

EDUCATION

Bachelor of Applied Science (BASc) | Manufacturing Engineering
University of British Columbia

Expected graduation: April 2027

PROFESSIONAL WORK EXPERIENCE

Composites Research Network Lab
Undergraduate Research Assistant

February 2025 - Present

- Lead an industry-partnered circular economy project focused on the selection of a recyclable, food grade material ensuring good heat transfer, moisture retention and 500°F+ thermal stability and performed tests to validate the selections (e.g. thermogravimetric analysis, prototype testing vs current product).
- Collaborated with a PhD student on research by creating a python program (using Open3D, NumPy, SciPy) with custom algorithms for automated processing and quantification of composite defects (e.g. fiber wrinkling, folds, and bridging) from 3D point cloud data.
- Designed a part replacement for a hot press device for in-house manufacturing, provided GD&T compliant CAD drawings, and collaborated with machine shop technicians for tolerance validation, eliminating external procurement lead times.
- Diagnosed injection molding device errors with minimal supervision, connected with OEM technical support and UBC engineering technician team to retrieve documents/drawings to review and create a repair plan.
- Curated and annotated large-scale datasets to optimize the training of a machine learning model.
- Authored a comprehensive technical protocol document (SOP) for a newly arrived torsion tester device.
- Analyzed materials testing data using Python libraries (NumPy, Pandas, Matplotlib) to generate technical visualizations for research publications.
- Co-authored a peer-reviewed paper on the moisture durability of Western Red Cedar <https://doi.org/10.3390/ma18184353>

ENGINEERING DESIGN TEAMS

UBCO Steel Bridge
Fabrication Team Lead

September 2025 – Present

- Organizing engaging and fun weekly meetings for the fabrication sub team.
- Train sub-team members the foundations of creating useable shop drawings in SolidWorks.
- Multidisciplinary collaboration with the structural team by translating their conceptual designs into high-fidelity SolidWorks models.
- Utilized Weldment and Sheetmetal tools to create precise 3D representations and automated cut lists to streamline manufacturing and assembly.
- Lead bridge joint design to meet annual competition specifications while ensuring structural stability and load-bearing integrity.

OK Motorsports FSAE
Vehicle Dynamics Engineer

September 2023 – August 2025

- Collected and analyzed performance data from track tests for improving and adjusting driver interface and vehicle dynamics
- Maintained work environment cleanliness to ensure safety and organization in the workshop
- Responsible for machining several parts for the race car in a machine shop, building experience with tolerancing and metrology.
- Designed a pedal box with GD&T compliant manufacturing drawings in Siemens NX for the driver training racing simulator.

TECHNICAL PROJECTS

Ember Alert | <https://youtu.be/prgUgCiNibc>

2024

- Awarded 1st Place Design in 2023-24 APSC 171 Design Competition
- Engineered an innovative solution to assist firefighters with finding underground embers post wildfires
- Conceived an all-in-one solution leveraging an ESP32 microcontroller hosting a web app to allow multiple users to access temperature and geolocation data being recorded by an advanced handheld sensor.
- Used SolidWorks to model enclosures, HTML, CSS, and JavaScript used for a custom web app for real-time data visualization.

AE86 SolidWorks Model | <https://youtu.be/9Xsc4sq8zzQ>

2024

- Produced a model of the Toyota AE86 with an emphasis on attention to accurate details, totaling to over 300 hours of work in SolidWorks.
- Final assembly incorporated over 20,000 parts including a working drive train, suspension, engine, and transmission.
- Developed comprehensive file management and assembly hierarchy rules to manage 400+ unique parts.

Laptop SSD Recycling

2023

- Developed a novel solution to address waste of aging or dead laptops by repurposing into USB mass storage devices
- Engaged in extensive use of Gantt software to model critical paths and risks, and assign tasks ensuring consistency of progress.
- Utilized OpenLCA to conduct a product life cycle analysis.
- Created weekly design reports made in Microsoft word on various stages of design process

CERTIFICATIONS

SOLIDWORKS Design Professional (CSWP)

Certificate ID: [C-6T2VRD8HP4](#)

SOLIDWORKS Design Associate (CSWA)

Certificate ID: [C-WASR2BPM6Y](#)

SOLIDWORKS Additive Manufacturing Associate

Certificate ID: [C-BREF4QHPXA](#)

RECOGNITIONS & AWARDS

Deputy Vice-Chancellor Scholarship for Continuing Students

2024, 2025

Dean's List

2024, 2025

1st Place Design 2023-24 APSC 171

2024

OTHER EXPERIENCE

Pacific National Exhibition

May 2023 - September 2024

Seasonal Attendant

- Managed high-volume logistics for a 3000-car daily parking facility and utilized conflict resolution skills and quick thinking to ensure positive guest experiences.

Science World

Summer 2022

Assistant Science Facilitator

- Managed displays and galleries and helped with opening and closing by ensuring areas were orderly and all items were orderly.
- Strengthened customer service skills by engaging with families and assisting visitors.

PROFESSIONAL ASSOCIATIONS & MEMBERSHIPS

Engineers and Geoscientists British Columbia (EGBC)

2023 – Present

SAE international | Collegiate Member (FSAE)

2023 – 2025

REFERENCES

Olivia H. Margoto

Lab Manager | Composites Research Network - Okanagan Node (CRNO)

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